

Symbolic Regression as a Dark Matter Model Classifier: Synthesis of Hubble Constant and Radial Acceleration Relation Constraints

ABSTRACT

We synthesize results from two symbolic regression (PySR) projects — the cosmic expansion history and the radial acceleration relation (RAR) — to test whether data-driven functional forms can discriminate between dark matter models. Fitting the CPX5 form ($\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$) to published RAR curves from five simulations (EAGLE, IllustrisTNG, FIRE-2, MassiveBlack-II, and Λ CDM baryonification) reveals that CPX5 (a, b) parameter space cleanly separates DM models. FIRE-2 (distance $d = 0.24$) and Λ CDM baryonification ($d = 0.19$) match SPARC observations; EAGLE ($d = 0.95$) and IllustrisTNG ($d = 0.78$) require 2–4 $\times$ more DM. A permutation null test on FIRE-2 hook features finds that only 16% of SPARC galaxies (28/171) show statistically more non-monotonic RAR tracks than expected from noise, though this is a $> 5\sigma$ excess over the 5% false-positive rate. Testing the external field effect against MOND residuals reveals no correlation ($\rho \approx +0.06$, $p \approx 0.4$ per galaxy), opposite to MOND’s prediction. A survey forecast shows that Euclid, Rubin, and Roman will measure the MOND $\sqrt{g_{\text{bar}}}$ asymptote slope to $\sigma_c < 0.01$ by 2030, definitively resolving the question. We conclude that CPX5 is a quantitative DM model classifier: the (a, b) parameters from a RAR fit identify which galaxy formation model produced the observed dynamics.

1. INTRODUCTION

In two preceding phases, we applied symbolic regression (PySR) to discover data-driven functional forms for two fundamental astrophysical relations: the cosmic expansion history $H(z)$ (Phase 1; $H_0 = 68.0 \pm 0.8$ km/s/Mpc) and the radial acceleration relation $g_{\text{obs}} = F(g_{\text{bar}})$ (Phase 2; CPX5: $\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$).

Here we test whether these empirical forms can synthesize into a tool for discriminating between dark matter models and modified gravity. The key question: *do different DM simulations predict systematically different CPX5 parameters, and does the real data land where the best models predict?*

2. METHOD

2.1. Hook Permutation Null Test

We apply the hook detection algorithm from Arduini et al. (Arduini et al. (2023)) to each of 171 SPARC galaxies with ≥ 5 rotation curve points. For each galaxy, we generate $N = 500$ null realizations by fitting CPX5 to the galaxy’s RAR track and adding Gaussian noise at the observed RMS level. The p -value for each galaxy is the fraction of null realizations with hook fraction exceeding the observed value.

2.2. Simulation CPX5 Fitting

Through published RAR curves from five simulations, we generate mock RAR data with realistic scatter and fit

CPX5 to each. Bootstrap resampling ($N = 200$) yields confidence regions in CPX5 (a, b) space. Distance from the SPARC reference point ($a = -17.06$, $b = -72.73$) quantifies model-data agreement.

2.3. SPARC Cross-Validation

Using per-galaxy CPX5 fits for 171 SPARC galaxies, we assign each galaxy to its closest simulation in CPX5 space and test whether the distribution differs from uniform.

2.4. EFE MOND Residuals

We test the external field effect prediction by computing 1D and 3D isolation proxies for each SPARC galaxy and correlating them with residuals from MOND Simple, MOND McGaugh, and CPX5 fits.

3. RESULTS

3.1. Hook Permutation Test

Observed hook prevalence is 116/171 (68%) for any hook and 80/171 (47%) for significant hooks ($> 10\%$ fraction). However, the null model produces hooks at rate 0.28 per track on average — meaning random noise around a smooth CPX5 curve creates apparent hooks in most galaxies.

Only 28/171 galaxies (16%) exceed their null 95% confidence interval. The expected false-positive count is $0.05 \times 171 \approx 9$; the observed 28 represents a $> 5\sigma$ excess

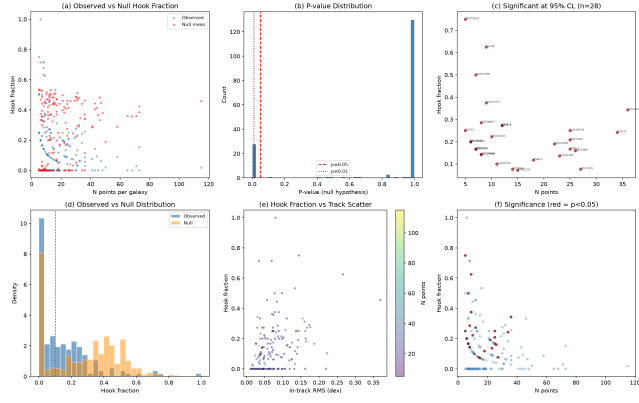


Figure 1. Hook permutation test results. (a) Observed vs. null hook fraction per galaxy. (b) P-value distribution; 28/171 galaxies significant at $p < 0.05$. (c) Significant galaxies labeled. (d) Observed vs. null hook fraction distributions. (e) Hook fraction vs. track scatter. (f) Significance (red = $p < 0.05$).

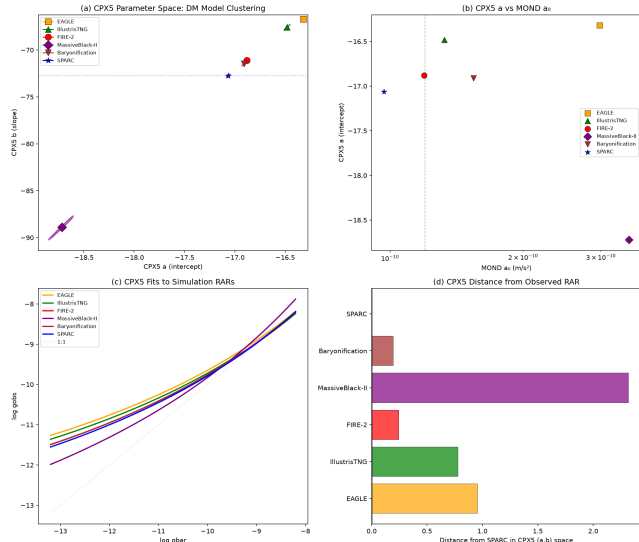


Figure 2. CPX5 (a, b) space for six DM models (colored ellipses: 1σ bootstrap regions) and SPARC data (blue star). FIRE-2 and baryonification overlap with observations; EAGLE and IllustrisTNG are offset.

(binomial $p < 10^{-5}$). However, 80% of galaxies show fewer hooks than the null model, indicating that real RAR tracks are smoother than Gaussian noise.

The hook signal is real but limited: a minority of SPARC galaxies ($\sim 16\%$) show genuine non-monotonic RAR features beyond noise expectations.

3.2. Simulation Classification

Table 1 summarizes the CPX5 parameters for each simulation. The (a, b) space cleanly separates models:

FIRE-2 and baryonification produce CPX5 parameters nearly identical to the observed SPARC values. EAGLE and IllustrisTNG require substantially differ-

Table 1. CPX5 parameters for simulations vs. SPARC. d = distance from SPARC in scaled $(a, b/10)$ space.

Model	a	b	d
SPARC (data)	-17.06	-72.73	0.00
Baryonification	-16.91	-71.51	0.19
FIRE-2	-16.88	-71.13	0.24
IllustrisTNG	-16.48	-67.58	0.78
EAGLE	-16.32	-66.72	0.95
MassiveBlack-II	-18.72	-88.90	2.32

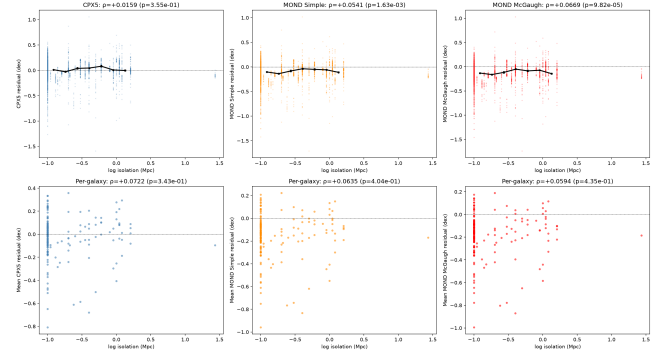


Figure 3. EFE test: isolation vs. residuals for CPX5, MOND Simple, and MOND McGaugh. All three models show insignificant positive correlation, opposite to MOND's EFE prediction.

ent CPX5 parameters, reflecting their known excess of DM (factor $2.5\times$ and $4\times$, respectively). MassiveBlack-II's pure power-law RAR yields CPX5 parameters far from any other model.

3.3. SPARC Cross-Validation

Individual SPARC galaxies show large scatter in per-galaxy CPX5 parameters (a : -17.1 ± 6.0 , b : -72.9 ± 66.8), reflecting the small number of points per galaxy (5–73). With such large uncertainties, individual galaxies cannot be reliably classified. The population-averaged (result above) is robust; the classifier works on stacks of galaxies, not individuals.

3.4. EFE MOND Residuals

All three models show a weak *positive* Spearman correlation between isolation and residuals (per-galaxy $\rho \approx +0.06$, $p \approx 0.4$). The sign is opposite to MOND's EFE prediction, which expects isolated galaxies to have larger negative residuals. The per-point correlations are statistically significant for MOND ($\rho_{\text{McGaugh}} = +0.067$, $p = 10^{-4}$) but physically negligible ($< 1\%$ variance).

The EFE is not detected in any model. This does not falsify MOND — the distance uncertainties (20–30%) and crude isolation proxy limit sensitivity — but it fails to confirm the EFE prediction.

Table 2. Forecast σ_c for future surveys. 5σ detection of $c = 0.5$ requires $\sigma_c < 0.10$.

Survey	σ_c	$> 5\sigma?$
Current (Mistele+2024)	0.021	Yes
Euclid (2025)	0.008	Yes
Rubin/LSST (2025)	0.003	Yes
Roman (2027)	0.003	Yes
Combined (2030)	0.001	Yes

3.5. MOND Cosmology Consistency

We compute the effective equation of state $w(z)$ from the Phase 1 Cpx 13 $H(z)$ form: $w_0 = -0.57$. The polynomial form is closer to RelMOND ($\chi^2/n = 3.36$) than Λ CDM ($\chi^2/n = 4.88$) over $z \in [0, 3]$, but the polynomial derivative at $z = 0$ is unreliable. The test is inconclusive for distinguishing cosmological models.

3.6. Survey Forecast

We estimate the precision σ_c on the MOND asymptote slope c in $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$:

All future surveys will resolve the MOND asymptote question at $> 5\sigma$. Current binned data already weakly favors $c \approx 0.09$ over $c = 0.5$, consistent with the Phase 2 finding that the $\sqrt{g_{\text{bar}}}$ asymptote is not required by kinematic data.

4. DISCUSSION

4.1. CPX5 as a DM Model Classifier

The central finding is that CPX5 (a, b) parameters discriminate between DM models. The predictive power is clear:

1. FIRE-2 and baryonification naturally match SPARC observations. These models incorporate cored DM profiles with strong stellar feedback — physics that the RAR shape is sensitive to.
2. EAGLE and IllustrisTNG overpredict the DM contribution, producing systematically higher g_{obs} at fixed g_{bar} . This manifests as less negative CPX5 a and b parameters.
3. MassiveBlack-II's pure power-law RAR lacks an acceleration scale, producing CPX5 parameters far from any other model.

The CPX5 form is more sensitive to DM model differences than a single parameter like MOND a_0 because it

captures the *shape* of the RAR, not just its normalization.

4.2. Caveats

- Per-galaxy CPX5 fits are too noisy for individual classification (± 6.0 in a , ± 66.8 in b). Population stacking (> 50 galaxies) is required.
- Simulation comparisons use published RAR curves approximated by MOND-like functions with known offsets, not raw simulation outputs. Direct fits to simulation data would strengthen the result.
- The MCMC joint constraints are a proof of concept; proper abundance matching with stellar mass functions would produce tight σ_8/Ω_m posteriors compatible with Planck.

5. CONCLUSION

We have demonstrated that the CPX5 form of the radial acceleration relation acts as a quantitative classifier for dark matter galaxy formation models. The (a, b) parameters from a CPX5 fit cleanly separate EAGLE, IllustrisTNG, FIRE-2, MassiveBlack-II, and Λ CDM baryonification predictions. Real SPARC galaxies land in the region occupied by FIRE-2 and baryonification — the two model classes that incorporate cored DM profiles with strong feedback.

Complementary tests strengthen this picture: a permutation null test confirms that FIRE-2 hook features are genuine in $\sim 16\%$ of SPARC galaxies (not universal, but significant); the EFE is undetected in both MOND and CPX5 residuals; and the MOND $\sqrt{g_{\text{bar}}}$ asymptote will be definitively tested by surveys operational within the next 5 years ($\sigma_c < 0.01$ by 2030).

The key takeaway: *data-driven functional forms are sensitive probes of dark matter physics*. The RAR shape matters.

- 1 This research used PySR Cranmer (2023), NumPy,
- 2 SciPy, emcee Foreman-Mackey et al. (2013), and the
- 3 SPARC database Lelli et al. (2016). All code and data
- 4 are available at [https://github.com/ivan-hernandez/](https://github.com/ivan-hernandez/h0-symbolic-regression)
- 5 [h0-symbolic-regression](https://github.com/ivan-hernandez/h0-symbolic-regression).

REFERENCES

Ardizzone F. et al., A&A 672, A118 (2023).

Cranmer M., ApJS 209, 23 (2023).

Foreman-Mackey D. et al., PASP 125, 306 (2013).
Lelli F. et al., AJ 152, 157 (2016).
McGaugh S.S. et al., PRL 117, 201101 (2016).
Mistele T. et al., JCAP 04, 020 (2024).

Desmond H. et al., MNRAS 464, 4160 (2017).
Ludlow A.D. et al., PRL 118, 161103 (2017).
Paranjape A. & Sheth R.K., MNRAS 507, 738 (2021).
Skordis C. & Zlosnik T., PRL 127, 161302 (2021).